Strategic Behavior

Midterm Exam • Spring 2024

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75 minutes • 60 points

Name: _____ Section / GSI: _____

Read this first.

- This exam has five questions. Each question tells a short story; your first job is always to translate the story into a game (players, strategies, payoffs).
- You may use one double-sided page of handwritten notes. A basic (non-graphing) calculator is permitted.
- Explain your reasoning in words as well as symbols. An answer such as "$q = 40$" with no derivation earns at most one point.
- Unless stated otherwise, players are risk neutral and maximize expected payoffs.

Question 1: The Cement Duopoly (14 points)

Two firms, Rocklin Aggregates (firm 1) and Tidewater Cement (firm 2), are the only suppliers of cement in a regional market. Each simultaneously chooses an output level $q_i \geq 0$ (in thousands of tons). The market price is

$$P(Q) = 120 - Q, \qquad Q = q_1 + q_2,$$

in dollars per ton (whenever $Q \leq 120$). Rocklin's newer kiln gives it a constant marginal cost of $c_1 = 12$; Tidewater's marginal cost is $c_2 = 24$. There are no fixed costs.

(a) *(4 pts)* Derive each firm's best-response function.

(b) *(5 pts)* Find the Nash equilibrium quantities, the market price, and each firm's profit.

(c) *(3 pts)* Holding $c_1 = 12$ fixed, for which values of c_2 does Tidewater produce a strictly positive quantity in equilibrium? What happens in the market when c_2 is above this threshold?

(d) *(2 pts)* Compute consumer surplus in the equilibrium of part (b).

Question 2: Where Should the Candidates Stand? (12 points)

Two candidates compete in a district whose voters have ideal policies spread over $[0,1]$ (0 is the most progressive position, 1 the most conservative). The district leans conservative: the voters' ideal points have density $f(x) = 2x$ on $[0,1]$, so the fraction of voters with ideal point below x is $F(x) = x^2$.

Each candidate simultaneously announces a platform $x_i \in [0,1]$. Every voter votes for the candidate whose platform is closest to her ideal point; if the platforms are equal, each candidate receives half of the votes. A candidate receives payoff 1 if she wins a strict majority, $\frac{1}{2}$ if the vote is tied, and 0 if she loses.

(a) *(3 pts)* Suppose $x_1 < x_2$. Write candidate 1's vote share as a function of x_1 and x_2.

(b) *(2 pts)* Find the median voter's ideal point m.

(c) *(3 pts)* Show that $(x_1, x_2) = (m, m)$ is a Nash equilibrium.

(d) *(4 pts)* Show that there is no other Nash equilibrium.

Question 3: Who Calls the Utility? (14 points)

During a storm a power line comes down on a quiet street. Each of the $n \geq 2$ households on the street sees it. If at least one household phones the utility company, the line is repaired and every household gains a benefit of $v = 12$. Phoning takes time and costs the caller $c = 3$. If nobody phones, every household gets 0. Households decide simultaneously and independently whether to phone.

(a) *(3 pts)* Find all pure-strategy Nash equilibria.

(b) *(5 pts)* Find the symmetric mixed-strategy Nash equilibrium in which each household phones with the same probability. Express the probability ρ that a given household does *not* phone as a function of n, c, and v.

(c) *(3 pts)* Compute ρ and the probability that nobody phones for $n = 2$ and $n = 3$.

(d) *(3 pts)* What happens to the probability that nobody phones as $n \to \infty$? Give the limit and a one-sentence interpretation.

Question 4: Ferry Pricing with Limited Seats (12 points)

Two ferry operators run the only crossings to an island. Each ferry has k seats per day and zero marginal cost. The daily number of passengers who wish to travel at price p is $D(p) = 100 - p$ for $0 \leq p \leq 100$. The operators set prices p_1, p_2 simultaneously.

Passengers buy from the cheaper ferry first. If $p_i < p_j$, operator i sells $\min\{k, D(p_i)\}$ seats, and operator j serves the residual demand: it sells $\min\{k,\ \max\{0, D(p_j) - k\}\}$ seats (the passengers who could not get a seat on the cheaper ferry are those with the highest willingness to pay). If $p_1 = p_2 = p$, each operator sells $\min\{k, D(p)/2\}$ seats. Each operator maximizes revenue.

(a) *(3 pts)* Let $k = 20$. Show that if both operators charge $p = 60$, both ferries are full, and compute each operator's revenue.

(b) *(5 pts)* Still with $k = 20$, show that $(p_1, p_2) = (60, 60)$ is a Nash equilibrium. Consider deviations both below and above 60.

(c) *(4 pts)* For general $k \in (0, 50)$, let $p^* = 100 - 2k$ be the price at which total capacity equals demand. Find all k for which (p^*, p^*) is a Nash equilibrium.

Question 5: Penalty Kicks (8 points)

In a penalty kick the kicker chooses to aim Left or Right and the goalkeeper simultaneously dives Left or Right. From tracking data, the probability of a goal is:

	Goalie dives L	Goalie dives R
Kicker aims L	0.55	0.95
Kicker aims R	0.90	0.60

The kicker maximizes and the goalie minimizes the probability of a goal.

(a) *(4 pts)* Find the mixed-strategy equilibrium and the equilibrium probability of a goal.

(b) *(4 pts)* After a season of practice, the kicker raises the scoring probability when aiming Right while the goalie dives Right from 0.60 to 0.70 (all other entries unchanged). Find the new equilibrium. Does the kicker now aim Right more or less often? Does the goalie dive Right more or less often? Explain the logic.

End of exam. Check that your name is on every page.